

Cybersecurity Threat Intelligence Report

Task-1: Awareness & Research Project

Internship Program : Cybersecurity Analyst Internship

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INTRODUCTION TO CYBERSECURITY

What is Cybersecurity?

Cybersecurity is the practice of protecting computers, networks, applications, and data from cyber attacks. It helps individuals and organizations keep their information safe from hackers and unauthorized access.

Why is Cybersecurity Important?

For Individuals:

- Protects personal information.
- Prevents identity theft.
- Reduces online fraud. **For**

Businesses:

- Protects customer data
- Prevents financial losses.
- Maintains company reputation.

Current Relevance:

- Increasing cybercrime worldwide.
- Growing dependence on digital technologies. Rise of cloud computing and IoT devices.
- AI-driven cyber attacks becoming more common.



THREAT 1 – AI-POWERED PHISHING ATTACKS

Description:

AI-powered phishing attacks use Artificial Intelligence to create realistic emails, messages, and voice calls that trick users into sharing sensitive information.

Impact:

Individuals

- Identity theft

- Financial fraud
- Account compromise

Organizations

- Data breaches
- Financial losses
- Reputation damage

Real-World Case Study:

In 2024, cybercriminals used AI voice cloning technology to imitate company executives and convince employees to transfer money to fake accounts.

Preventive Measures:

- Enable Multi-Factor Authentication (MFA)
- Verify requests before sharing information
- Conduct employee awareness training.



THREAT 2 – RANSOMWARE-AS-A-SERVICE (RaaS)

Description:

Ransomware-as-a-Service allows cybercriminals to rent ransomware tools and launch attacks without advanced technical skills.

Impact:

Individuals

- Loss of personal files.
- Financial loss.

Organizations

- Business downtime.
- Data loss.
- Recovery expenses.

Real-World Case Study:

The WannaCry attack infected more than 200,000 computers across 150 countries and caused major disruptions.

Preventive Measures:

- Maintain regular backups
- Update software regularly.
- Use antivirus protection



THREAT 3 – CLOUD SECURITY MISCONFIGURATIONS

Description:

Cloud misconfigurations occur when cloud storage or services are not configured properly, exposing sensitive information.

Impact:

Individuals

- Data exposure.
- Privacy risks.

Organizations

- Data breaches.
- Compliance issues. • Financial penalties.

Real-World Case Study:

The Capital One breach in 2019 exposed customer information because of a cloud configuration issue.

Preventive Measures:

- Perform regular security audits.
- Apply least-privilege access.
- Monitor cloud environments.



THREAT 4 – IoT VULNERABILITIES

Description:

IoT devices such as smart cameras, smart TVs, and smart home devices may contain security weaknesses that hackers can exploit.

Impact:

Individuals

- Privacy invasion.
- Device misuse.

Organizations

- ✦ Unauthorized network access
- ✦ Security breaches.

Real-World Case Study:

The Mirai Botnet attack infected thousands of IoT devices and used them to launch large-scale cyber attacks.

Preventive Measures:

- Change default passwords.
- Update device firmware.
- Secure Wi-Fi networks.



THREAT 5 – ZERO-DAY EXPLOITS

Description:

A Zero-Day exploit targets software vulnerabilities before developers release a security patch.

Impact:

Individuals

- Device compromise
- Personal data theft.

Organizations

- Data breaches
- Service

disruption.

Real-World Case Study:

The Microsoft Exchange Server attack exploited unknown vulnerabilities and affected thousands of organizations.

Preventive Measures:

- ✦ Install updates promptly.
- ✦ Use intrusion detection systems.
- ✦ Monitor network activity continuously.



CONCLUSION

Conclusion:

- Cybersecurity is essential for protecting digital information and systems from modern cyber threats. AI-powered phishing, ransomware, cloud security issues, IoT vulnerabilities, and zero-day exploits continue to evolve and create new challenges.
- Organizations and individuals must adopt proactive security measures such as Multi-Factor Authentication, regular software updates, security awareness training, and continuous monitoring. Staying informed and continuously learning about emerging threats is critical for maintaining a secure digital environment.

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